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Mapping industrial lobbying strategies in the EU PFAS restriction using AI-assisted text analysis

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ABSTRACT

The European Union's proposed restriction of per- and polyfluoroalkyl substances (PFAS) prompted unprecedented stakeholder engagement. This study maps industrial lobbying strategies within the 2023 European Chemicals Agency public consultation, the largest in EU history. Combining manual qualitative coding with large scale AI-assisted classification, we analysed stance, exemption requests, and argumentation patterns across 4771 unique comments.

The analysis reveals a dominant strategy among industrial actors, who contributed 68.5% of all submissions. While 71% of industry stakeholders explicitly opposed the restriction, 78% of these opponents simultaneously requested technical exemptions. This approach contests the regulation's legitimacy while ensuring technical derogations to weaken its scope. Argumentation clustered around four narratives: economic competitiveness, essentiality, scientific minimization, and responsible innovation. Beyond these empirical findings, we provide an open-access sectoral dashboard to facilitate regulatory oversight and visualize these lobbying patterns. Our analysis reveals an asymmetry in lobbying power by industries that rely heavily on PFAS functionality compared to other stakeholders. The results demonstrate how technical complexity and mass participation can be strategically leveraged to influence regulatory ambition and reframe a debate on chemical risks and public health, into one centered on European competitiveness and technological indispensability.

1. Introduction

Industrial lobbying has historically played a central role in shaping the regulation of hazardous chemicals. Well documented cases such as dichloro-diphenyl-trichloroethane (DDT), tobacco and glyphosate illustrate how industries have repeatedly employed tactics to delay restrictions, question scientific evidence, emphasize economic costs, and frame products as essential (Kaurov and Oreskes, 2025; Reed et al., 2021; Michaels and Michaels, 2020; Wurster, 2015).

To move beyond historical observation toward a structural understanding of these influences, researchers utilize frameworks such as the Science for Profit Model (SPM) (Legg et al., 2024). Originally developed to analyse corporate influence in the tobacco industry, the SPM is a typology designed framework to detect corporate influence on science and policy, specifically identifying strategies for the deliberate manufacturing of ignorance or doubt (agnogenesis) (Green et al., 2025; Legg et al., 2024, 2021). Similar patterns of challenging regulatory evidence have been documented across other sectors, such as the alcohol industry, to shift the burden of proof away from the state (Savell et al., 2016).

The Forever Lobbying Project has recently highlighted how these tactics recur across chemical regulation debates, including in the case of per- and polyfluoroalkyl substances (PFAS), a group of highly persistent synthetic compounds (Horel and Aubert, 2025). PFAS are defined by the presence of at least one fully fluorinated carbon atom (–CF_2– or –CF_3) (OECD, 2021). Based on this broad definition, thousands of individual PFAS are currently present on the global market, and extrapolations from large chemical databases such as PubChem even suggest that the number could reach into the millions (Schymanski et al., 2023), and the governance challenges surrounding their regulation are equally complex. Their unique physicochemical properties, including extreme stability, resistance to degradation, and amphiphilic behaviour, have

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driven their widespread use in industrial processes and consumer products, including textiles, food packaging, electronics, and fire-fighting foams (Glüge et al., 2020). At the same time, these properties contribute to environmental persistence, long range mobility, and bio-accumulation in humans and wildlife (Camoiras González et al., 2021; Fiedler et al., 2022; Fiedler and Sadia, 2021; Gkika et al., 2024; Sadia et al., 2023). Exposure to PFAS has been associated with developmental, endocrine, immunological and carcinogenic effects (Bline et al., 2024; Mayilswami et al., 2025).

In 2023, five European countries (the Netherlands, Germany, Norway, Sweden, and Denmark) submitted a joint proposal to restrict PFAS under the EU REACH regulation (ECHA, 2023a). The proposal adopts the OECD PFAS definition and seeks to phase out nearly all uses of the class within this REACH context, while allowing extended transition periods for critical sectors such as medical devices. Uses not regulated by the REACH regulation, such as pesticides or pharmaceuticals (comprising roughly 45 active pesticide ingredients and around 342 pharmaceutical substances meeting the OECD definition of PFAS (Brunn et al., 2023), are not included in this restriction proposal. During a six-month public consultation (March–September 2023), over 5,600 comments were submitted by stakeholders which were in majority industries, alongside non-governmental organisations, governments and individual citizens. This submission volume is nearly twelve times higher compared to the previous record for a restriction proposal on microplastics (ECHA, 2024; KEMI, 2023).

The high volume of responses underscores the socio-economic stakes of this restriction proposal. Furthermore, it highlights a critical vulnerability in participatory governance: while public consultations are designed to democratize policy making, they can be disproportionately dominated by well-resourced corporate actors. The intensity of regulatory and industry engagement contrasts with the still relatively low public awareness regarding PFAS, even in countries directly affected by contamination (Berthold et al., 2023; Gary et al., 2023). For example, despite the National Institute for Public Health and the Environment (RIVM) in The Netherlands advising against consumption of eggs from hobby chickens due to PFAS, many inhabitants continued consumptions, highlighting knowledge gaps (NU.nl, 2025a, 2025b). Across the EU, civil society organizations have amplified the debate, but social responses remain fragmented. This highlights the gap between scientific evidence, regulatory processes, and public concern, creating fertile ground for lobbying strategies.

The aforementioned six-month public consultation of the PFAS restriction proposal provides a unique opportunity to systematically map the lobbying strategies. To systematically identify and characterize these lobbying arguments, this study combines manual qualitative coding with large scale artificial intelligence (AI) assisted classification of consultation comments. For the latter we used a simplified framework adapted from the Science for Profit Model (SPM) (Legg et al., 2021), a tool that enables the detection of systematic industry bias within a massive corpus of technical comments. Our primary objectives were twofold: to map lobbying strategies employed by PFAS producing industries and to evaluate the effectiveness of the AI-assisted tool in detecting industry bias in regulatory discourse. The study thereby aims to enhance regulatory transparency and to inform resilient environmental policy.

2. Materials and methods

2.1. Data sources and extraction

The dataset consisted of 5642 individual comments submitted to ECHA during the March–September 2023 public consultation on the proposed PFAS restriction. Each record contained the full text of the comment, submitter metadata (e.g., organization type, country, request for exemption, and sector) and information on attachments (ECHA, 2023a, b).

Of these 5642 comments, 1876 included an attachment, the majority of which (65%, n = 1226) were flagged as confidential under ECHA's confidential business information guidelines, making them legally and technically inaccessible for public research. All attachments were excluded from further automated text analysis because their variable formats (often non-machine-readable scans) and extreme variations in length would have compromised computational comparability and significantly increased processing costs. Crucially, this exclusion applied only to the attached files, not to the submissions themselves. While organized industry consortia were expected to submit comprehensive statements as attachments, they typically populated the online text boxes with cover text or summaries as advised by ECHA portal, meaning that their perspectives remained fully represented in our main analytical corpus. To empirically validate this assumption and control for potential selection bias, we conducted a manual qualitative audit of a random sample of 35 out of the 650 publicly available, non-confidential attachments. This targeted check revealed that these files primarily comprised supplementary technical literature, corporate socioeconomic worksheets, and detailed technical exemption applications rather than novel macro level policy arguments. This confirms that the online text boxes effectively captured the submitters' core regulatory stances and primary lines of argumentation, demonstrating that the technical exclusion of attachments from the automated process does not systematically distort our sector level findings. However, excluding these attachments stands as an analytical limitation.

To ensure data quality and analytical focus, a two-stage filtering process was applied. First, identical duplicate submissions from the same organizations were removed to avoid skewing results toward high volume commenters. Second, submissions were excluded because they lacked textual content in the public comment fields (e.g., containing only symbols like "-" or references to missing text). In total, 871 submissions were removed. Consequently, the final analytical dataset consisted of 4771 unique submissions with complete stance and argument mapping.

To ensure comparability across languages, 883 non-English submissions were translated into English using OpenAI's GPT-3.5 model, which has been validated for regulatory text translation (Ajai Krishna et al., 2024). Metadata harmonisation was conducted by assigning organisations to the Statistical Classification of Economic Activities in the European Community (NACE) classification system to enable sector level aggregation (Ulrich et al., 2025; van Wezel et al., 2018; Eurostat, 2008). Assignment uncertainty was limited at aggregated sector levels and primarily affected organisations with diversified or cross sector activities; such cases were conservatively coded at higher NACE levels to minimise misclassification bias (Eurostat, 2008).

2.2. Analytical framework

To systematically analyse lobbying discourse in the consultation, we focused on two primary complementary dimensions: stance and argumentation, both detailed below. These two dimensions were applied consistently to both a manually coded Dutch subset and a large scale AI-assisted classification of the full EU dataset to ensure comparability. Furthermore, the AI-assisted analysis enabled a third macro-level dimension regarding the sectoral distribution. By aggregating classified stances and arguments across industrial sectors, we were able to identify broader lobbying patterns across the European economic landscape that were beyond the scope of the manual subset.

2.2.1. Stance categories

Stance analysis was implemented to provide an overview of stakeholder alignment toward the proposed regulation. Each comment was categorized by stance, reflecting the position of the submitter towards the proposed restriction: 1) support, 2) oppose, 3) neutral. Operationally, 'Neutral' was defined as the absence of an explicit, one sided regulatory position. This three part classification aligns with categories

commonly used in policy consultation analyses (Bunea, 2017; European Commission, 2019).

Because stakeholders often adopt a nuanced position, for example by supporting the restriction in principle while seeking a specific derogation, we also utilized the "Request for exemption" label provided within the ECHA consultation dataset. This information was extracted from the "Content" column of the raw data as a cross cutting strategic dimension. In our framework, the exemption request is treated as a specific lobbying objective that can be paired with any stance. This allows for the detection of "conditional support" (supporting the ban while asking for an exemption), "strategic opposition" (opposing the ban while requesting exemptions as a secondary fallback), or a "technical-neutral position" (where a stakeholder requests an exemption for a specific use without taking a clear political stance on the overall restriction).

2.2.2. Argument taxonomy

Argument analysis was implemented to capture the narratives and claims used by stakeholders to justify their stance toward the proposed PFAS restriction. A structured taxonomy was developed to classify recurring argument types, ranging from scientific claims about PFAS hazards to anticipated economic or societal implications of regulation. The taxonomy builds conceptually on the lobbying narratives identified by the Forever Lobbying Project (Horel et al., 2025; Horel and Aubert, 2025), which documents a broad set of recurring corporate argumentation strategies in the context of PFAS regulation. In that project lobbying arguments are presented as a comprehensive descriptive scheme, encompassing a wide range of fine grained claims, premises, and associated lobbying requests.

In the present study, this descriptive scheme was explicitly operationalised and streamlined for the purposes of systematic text analysis. During exploratory scheme development, a broad set of argument claims described in the Forever Lobbying Project (Horel et al., 2025; Horel and Aubert, 2025) was initially considered. While several claims appeared only in isolated instances, others were consolidated to ensure high AI classification accuracy and analytical tractability. Specifically, narrow technical claims, such as those regarding the sufficiency of persistence as a ground for restriction, were consolidated into broader scientific or regulatory narratives in order to allow to more reliably identify when stakeholders were challenging the scientific justification of the restriction. As this refinement occurred before systematic coding, the process was qualitative rather than frequency driven. Closely related claims were grouped under higher level argument categories to ensure analytical tractability, consistency in manual annotation, and robustness in AI-assisted classification. Methodologically, the 'Scientific' category captured these specific technical and hazard profile premises as defined in Table 1, irrespective of whether formal academic citations were appended to the submission.

While the Forever Lobbying Project distinguishes primarily between scientific, "no alternative", and economic arguments, two additional categories were introduced. First, a separate category of societal arguments was introduced to capture claims related to impacts on public services, daily life, health systems, and the green or digital transition, which recurred across submissions but were not explicitly formalised as a stand alone group in the original scheme. This category also includes broader policy level arguments, such as the application of the precautionary principle and alignment with EU Green Deal objectives. Second, a residual "No arguments" category was included to account for comments that did not contain discernible lobbying argumentation.

The final taxonomy comprised twelve sub-arguments grouped into five overarching categories, to be exhaustive for the purposes of classification and to ensure that all consultation comments could be assigned to at least one category (Table 1). This taxonomy was implemented consistently in both manual coding and AI prompting, allowing individual comments to be assigned to multiple argument types where applicable.

2.2.3. Manual comment analysis

In total, 117 comments were submitted from the Netherlands, these were manually annotated to provide a baseline for comparison and contextual depth. Each comment was coded for stance, and those expressing opposition or seeking exemptions were further analysed using the argument taxonomy. This manually annotated subset served to validate the technical performance of the AI-assisted classification (stance and multi-label arguments), and to provide a context specific qualitative illustration of lobbying patterns.

2.2.4. AI-assisted comment analysis

Large scale automated analysis was conducted using large language models (LLMs) and natural language processing (NLP) techniques. The full dataset of 4771 comments was analysed for stance detection and argument classification. The outputs were aggregated at the sector level to assess patterns across stakeholder groups and economic sectors.

For argument analysis, the primary task was to classify argument taxonomy according to Table 1. A three-run ensemble of GPT-4o-mini with zero-shot prompting (Pietron et al., 2024; Yin et al., 2019) was adopted as the main configuration. This setup was chosen after testing alternative models (e.g., GPT-4o, RoBERTa) and prompt engineering techniques (e.g. few-shot, chain-of-thought). The GPT-4o-mini ensemble offered the best trade-off between classification accuracy and computational efficiency. Each comment was processed three times, with labels assigned by majority vote to increase stability.

For stance detection, several prompting strategies were evaluated, including zero-shot, one-shot, and few-shot approaches. A few-shot configuration using nine manually labelled examples (three per stance category) was selected based on agreement with manual coding. By providing the model with a structured representation of each stance, this approach reduced misclassification, particularly between opposing and exemption seeking comments.

For sector level sentiment analysis, sector specific lobbying patterns were examined by aggregating stance and argument classifications

Table 1
Main groups of lobbying argument taxonomy.

Argument	Label	Claim
Scientific	Polymer of low concern	A given PFAS polymer is less dangerous.
	Negligible toxicity	A given PFAS chemical is less toxic.
No alternative*	No alternative	There is no alternative to replace a given PFAS.
	Alternatives underperform	Existing alternatives perform worse than a given PFAS based material.
Economic*	Economic impact	There is a negative economic impact on business.
	Forced relocation	Production of given PFAS and PFAS based materials is forced to move out of Europe.
	Impact on jobs	Job loss because of the PFAS restriction.
	Competitiveness loss	Loss of EU competitiveness and strategic autonomy.
Societal	Customer convenience	Negative impact on customer convenience and modern life after losing given PFAS based products.
	Health and safety	Negative impact on the health sector after loss of given PFAS-based products.
	Green transition	A given PFAS reduces greenhouse gas or process related emissions.
No arguments	No arguments	No discernible lobbying argumentation.

* represent argument adapted from Forever lobbying project.

across industrial sectors. Zero-shot classification using BART and GPT-based prompting (GPT-3.5) were applied to identify dominant sentiment orientations and exemption focused argumentation within sectors.

To ensure the reliability of the automated classifications, the AI models were again validated against a set of random collected comments (100 comments for argument analysis, 250 comments of stance analysis), which were manually annotated. The model's accuracy was primarily measured using the F1 score, a standard metric in natural language processing that provides a balanced measure of accuracy (Malyi et al., 2026; Vickers et al., 2023; Walsh et al., 2024). Detailed performance metrics for every category are documented in the Supplemental Information (Section S1, Tables S1–S3).

The models achieved strong performance on major classification tasks (Tables S1–S3), with macro F1-scores of 0.87 for stance detection and 0.91 for grouped lobbying arguments. However, performance varied across fine grained subcategories due to limited sample sizes and conceptual overlap. Notably, argument label 1.2 ('Negligible toxicity') yielded an F1-score of 0, resulting from a small validation sample (n = 9) a and the inherent difficulty of the language model in distinguishing highly specific, sparse technical claims from other broad scientific arguments within the dataset. While this limits the model's precision in separating granular scientific nuances, confidence in the primary downstream findings remains high. The primary lobbying dynamics identified in this study rely strictly on stance classification (F1 = 0.87) and the primary structural arguments regarding 'No Alternatives' (F1 = 0.95) and 'Economic' impacts (F1 = 0.86), both of which demonstrate robust classification reliability. Finally, given the rapid pace of development in LLM frontiers, the performance metrics reported in this study may represent a conservative baseline; utilizing newer generation models within this same classification process would likely yield higher overall accuracy, particularly for highly specific categories.

All scripts, prompts, and configuration files used for preprocessing, argument classification, stance detection, and sectoral analysis are available in an open access GitHub repository, see the Section S2 in SI. In addition, the complete text of all engineering prompts, including exact system instructions and few-shot examples used for stance and argument classification, is explicitly detailed in the Section S3 in SI.

Throughout this study, the term 'AI-assisted analysis' refers specifically to a Large LLM assisted classification and coding used to process, label, and structure the raw consultation comments at scale. Primarily, the LLM's role was strictly limited to text categorization based on predetermined taxonomies; it was not used for numerical calculations, or statistical analysis, which were performed entirely and conventionally by the authors.

3. Results and discussion

3.1. Stakeholder engagement in the PFAS consultation

For the 2023 ECHA consultation on the proposed PFAS restriction, industries were by far the most active contributors, while individuals, NGOs, governmental authorities and academia comprised the remainder (Fig. 1). Contributions came from 52 countries, with the largest shares originating from Sweden, Germany, and Japan, which together accounted for nearly two-thirds of all comments. The prominence of non-EU submissions, notably from the United States, Japan, and China, demonstrates the significant global relevance that the proposed EU restriction has for PFAS related supply chains and trade (see Fig. S1 for a global distribution map).

The dominance of industry submissions underscores an asymmetry in stakeholder engagement: corporate actors appear highly organized and coordinated as compared to the fragmented participation from civil society and individuals. An exception was observed in Sweden, where over 1,300 individual submissions were recorded. This spike in public participation might be explained by an outreach campaign coordinated by the Swedish Society for Nature Conservation (Naturskyddsföreningen, 2024) to raise awareness and to facilitate citizen input on this consultation on the proposed PFAS restriction.

3.2. Manual analysis of Dutch comments

To provide cultural and contextual depth, a subset of 117 Dutch comments was manually analysed. In agreement with the full dataset, the majority of Dutch submissions (79%) originated from industrial and manufacturing sectors. The stance distribution across the 117 comments was relatively balanced: 39% supported the restriction, 30% expressed opposition and 31% remained neutral. Strategic exemption seeking was however high across all three groups, at 43%, 69%, and 67%

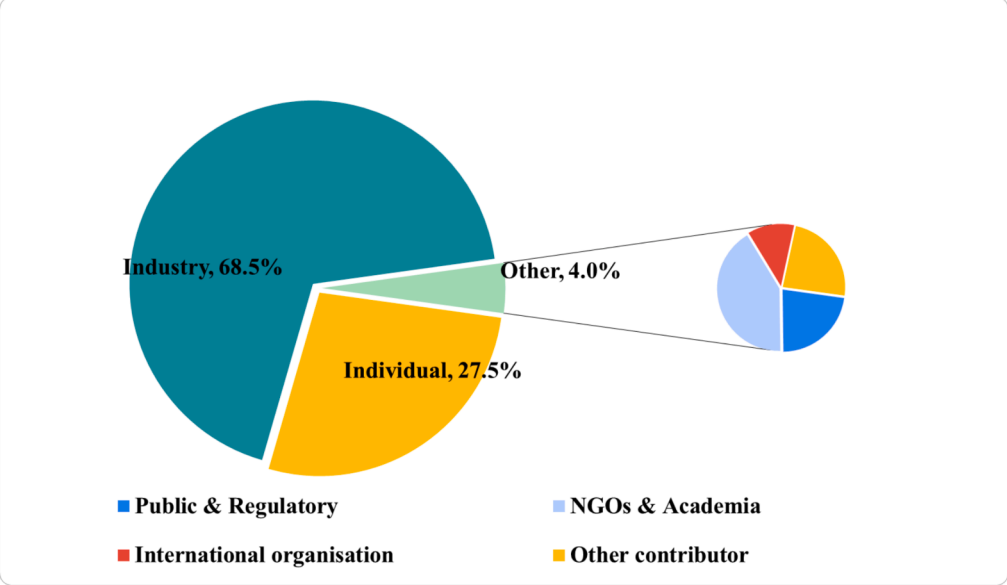


Fig. 1. Stakeholder distribution in the 2023 ECHA PFAS consultation. Categories are derived strictly from official ECHA submitter metadata. Industry includes individual companies and trade associations; Individual represents private citizens; Public & Regulatory consolidate national authorities; NGOs & Academia includes civil society and research organizations.

respectively (Fig. S2).

Comments from industry actors, primarily representing sectors such as semiconductor, chemical and industrial manufacturing (classified via NACE coding), typically cited economic and technical arguments (Fig. S3). These submissions emphasized the lack of viable alternatives and potential economic harm. As example, a Dutch SME in the conveyor belt sector (NACE C28) argued that a ban without exemptions would force a complete company closure and the loss of over 50 jobs, as no functional alternatives to PTFE coated belts exist.

In contrast, comments from individuals, NGOs, authorities, and other contributors, though fewer in number, were more supportive to the restriction. These submissions often invoked scientific and societal arguments, specifically citing PFAS toxicity and long term health impacts. For instance, Dutch regional authorities (Provinces) explicitly referred to the 'precautionary principle' and the need for alignment with EU Green Deal objectives, highlighting the widespread contamination of Dutch groundwater and soil as a primary driver for a total ban.

A subset of comments exhibited mixed positions, supporting the general phase out of PFAS while simultaneously requesting sector specific exemptions (Fig. S2). This hybrid stance illustrates how actors may simultaneously recognize environmental risks while defending particular economic interests. Notably, a coordinated submission effort was

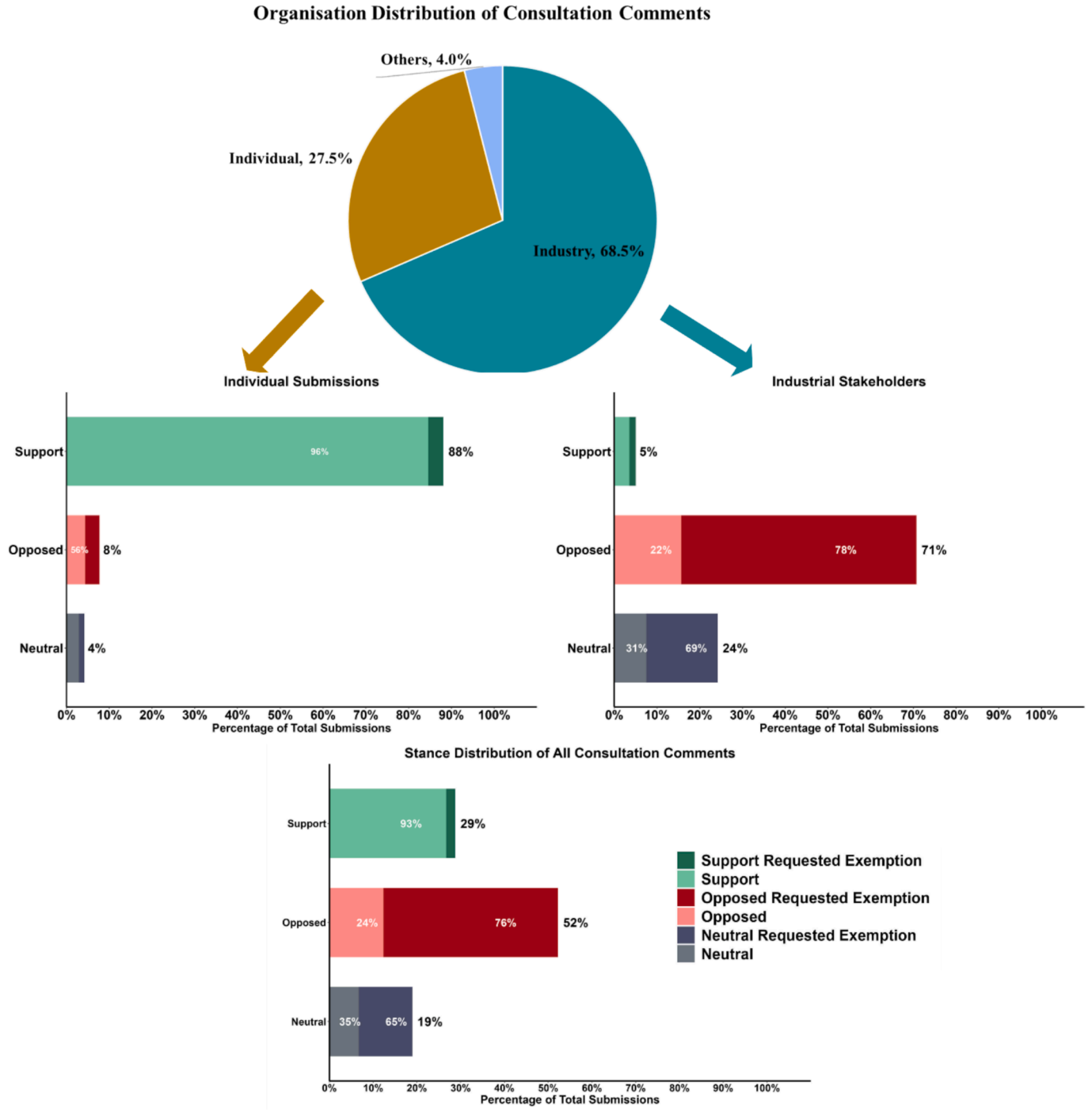


Fig. 2. The distribution of all consultations by organization type and by stance. The upper pie chart illustrates a clear structural asymmetry in stakeholder participation. The lower breakdown charts reveal highly polarized stakeholder orientations. The dark shaded segments highlight a widespread strategic reliance on derogations (comments are coupled with explicit requests for technical exemptions).

identified among several municipalities in the Zuid-Holland region, specifically Dordrecht, Papendrecht, Sliedrecht, Molenlanden, and Alblasserdam. A cluster of 13 submissions shared nearly identical text, emphasizing the local burden of environmental contamination (PFAS producer Chemours is located in Dordrecht) while calling for a balanced regulatory approach.

In addition to stance, an argument type analysis was conducted on 57 Dutch comments classified as "oppose" or "opposed requested exemption" (Fig. S3). Industrial and nonindustrial stakeholders relied on similar argument categories. The "no alternative" argument clearly dominates, followed by "health and safety" arguments. More than half of all analysed comment, across both companies and other stakeholders claimed that no viable alternatives currently exist to replace PFAS in critical applications, often accompanied by secondary claims emphasizing the inferior performance of alternatives and potential negative business or competitiveness impacts. Health and safety arguments formed the second most frequent category, raised by both companies and other stakeholders to justify continued PFAS use, often citing the reliability and performance of PFAS based materials in protective equipment, or medical applications. Economic arguments, including negative business impact, job loss, and loss of competitiveness, appeared in roughly one-third of company submissions, framing the proposed restriction as a potential threat to European industry and innovation. Scientific arguments, such as polymer of low concern (PLC) and negligible toxicity, were limited in number and primarily raised by company respondents. Societal arguments such as the green transition or convenience were not dominant amongst either group.

3.3. AI-assisted large scale comment analysis

The AI-assisted large scale comment analysis framework was applied to the full dataset of 4771 unique comments submitted to the 2023 ECHA PFAS restriction consultation. All analytical outputs are available through a simplified interactive online dashboard (SI, Section S2) that visualizes stance, argument, and sectoral patterns across the dataset, where it is possible to explore how positions vary across countries, organization types, and argument categories. The classification achieved high technical validation (SI, Section S1) and confirmed the general trends as observed in the Dutch subset, while providing a more detailed cross sectoral perspective.

The results show that opposition and exemption seeking are the dominant features of the discourse (Fig. 2). Overall, 52% of comments were classified as opposing the restriction, while 29% were supportive and 19% remained neutral. The analysis reveals a high reliance on exemptions, particularly among those resistant to the ban. An overwhelming 76% of opposing comments were paired with specific requests for exemptions, suggesting that stakeholders attempt to fragment the universal ban into a for them more manageable and less restrictive regulatory framework, similar to the restriction proposal on microplastics (Clausen et al., 2020). In contrast, only 7% of supportive comments requested exemptions, indicating that supporters generally advocate for a comprehensive ban without significant loopholes. Notably, over half of the neutral submissions (65%) also requested exemptions, reflecting a "technical neutral" strategy where actors focus on securing specific uses rather than engaging in the broader policy debate.

The majority of these exemption seeking comments originated from industrial actors, particularly within the chemical (n = 704, with 72% seeking exemptions), electronics (n = 473, with 79% seeking exemptions), and manufacturing sectors (n = 378, with 78% seeking exemptions). This underscores the continued dominance of industrial interests in shaping the technical complexity of the consultation through a high volume of specific requests. In contrast, comments from nonindustrial stakeholders, such as NGOs and governmental authorities, were less numerous and significantly more likely to advocate for a comprehensive restriction without exceptions (Fig. 2). Notably, while individual citizens generally expressed support for the restriction, 39% of these supportive individual comments included requests for exemptions. This suggests that even among those who broadly support the environmental goals of the ban, there is a simultaneous desire to protect specific personal or technical uses deemed essential.

The argument type classification identified systematic lobbying narratives, which are consistent with patterns observed in historical analyses of pesticide regulation (e.g., DDT) and broader studies of corporate political activity across health- and environment-relevant industries (Conis, 2016; Ulucanlar et al., 2023). Across all comments submitted by industries, the "no alternative" argument group was used most frequently, serving as the primary pillar of industrial resistance (Fig. 3). This claim is present in the vast majority of opposing submissions (86%), framing PFAS not merely as a choice, but as an irreplaceable technical necessity for modern industry, such as in medical devices, semiconductors, or protective coatings.

This strategy creates a logical "cascade" for subsequent arguments: first establishing that no alternatives exist; then presenting an economic threat and a ban would lead to unavoidable business disruption (71%) and to loss of European competitiveness (60%), this then leads to societal consequences, such as a threat to the green transition (27%) and health & safety (19%), as essential technologies would supposedly disappear. This narrative structure effectively shifts the debate from a political choice about environmental safety to a technical impossibility. Furthermore, the high frequency of scientific arguments, specifically "polymers of low concern" at 34% among opponents, suggests a secondary narrative: if a total ban cannot be avoided through the "no alternative" claim, the industry seeks to narrow the regulation's scope by exempting specific chemical categories. While exploring the precise technical boundaries between these chemical subgroups (such as fluoropolymers versus non-polymeric PFAS) is outside the scope of this broad text analysis framework, it represents an important avenue for future research using specialized taxonomies.

In contrast, the narratives used by supporters and neutral stakeholders are significantly less developed, focusing more on general policy alignment than technical properties. While unsurprising, given that technical expertise is heavily concentrated within industry, and the defense of the status quo inherently demands granular technical justification. This disparity underscores a "narrative dominance" by industrial actors, who leverage technical complexity to justify the continued use of forever chemicals.

By mapping the argument taxonomy (Table 1, Fig. 3) onto the Science for Profit Model (SPM), the argument type classification identified four dominant rhetorical clusters that align with the SPM (Table S2). The most prevalent is economic and regulatory resistance, characterised by claims of economic impact, forced relocation, and job loss, reframing the restriction as a fiscal threat rather than an environmental necessity. This is reinforced by essentiality and irreplaceability framing, where "no alternative" and health-and-safety claims portray PFAS as technologically indispensable. A third cluster, scientific minimization, operates through selective use of "polymer of low concern" and "negligible toxicity" labels to narrow the regulation's scope, echoing well documented strategies of agnogenesis (Van Zee, 2009). Finally, a responsible innovation narrative advocates for voluntary phase out and recycling to portray industry as a cooperative partner while retaining control over the regulatory timeline (Gaber et al., 2023; Oreskes and Conway, 2010). Together, these clusters reveal a coherent rhetorical system in which industrial actors synthesize economic threat, scientific doubt, and responsible innovation to transform the discourse. This synthesis reinforces how PFAS lobbying mirrors historical corporate strategies to delay, deflect, and domesticate regulation (Gaber et al., 2023; Oreskes and Conway, 2010; Saltelli et al., 2022; Schäffer et al., 2023).

PFAS producing and PFAS dependent sectors are the primary sources of regulatory resistance, characterized by a high volume of exemption requests. Sector specific further analysis (see Fig. 4) shows that in key sectors such as chemicals, electronics, and machinery, the demand for exemptions was considerably more prevalent than comments opposing

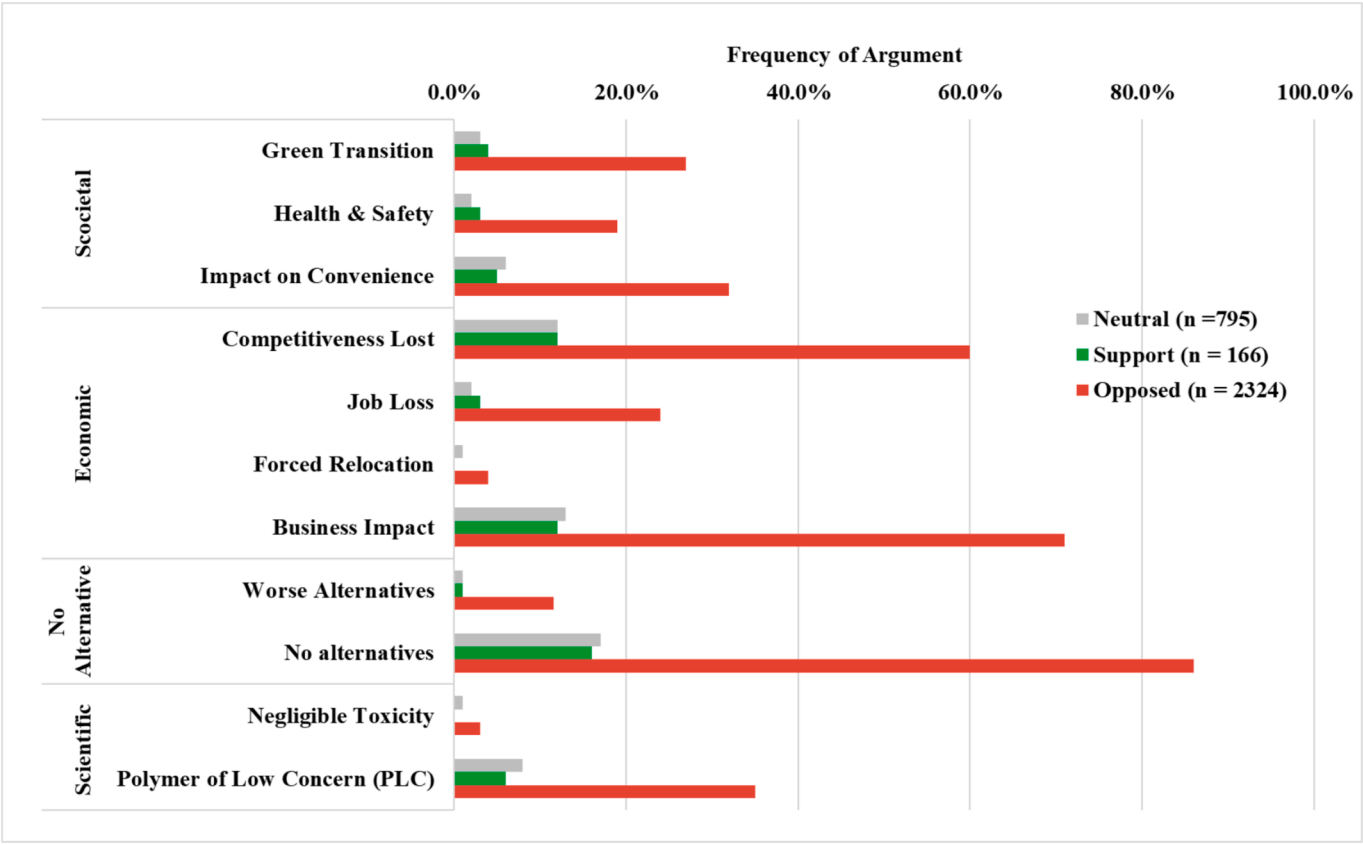


Fig. 3. Distribution frequency of argument types (the scientific, no alternative, economic, and societal categories) across industrial stances, broken down by opposing, supportive, and neutral industry positions (n = 3285).

the restriction entirely. Specifically, these three sectors, alongside rubber and plastic products, accounted for over 60% of both total opposition and total exemption demands. This demonstrates that PFAS dependent industries are not merely resisting the regulation but are strategically leveraging 'no alternative' and 'negative economic impact' arguments of specific PFAS to secure long term derogations. Smaller clusters of combined opposition and exemption requests were also observed in metal production, transport, and pharmaceuticals, indicating that the demand for continued PFAS use is embedded deep within global value chains. Further details for each specific industry and stance can be extracted from the interactive online dashboard (SI, Section S2).

In contrast, supportive comments were largely associated with individual submissions (n = 1150) and nonindustrial actors (n = 166). Industrial support (n = 166) for the restriction was limited and originated primarily from the chemicals (n = 24), machinery (n = 17) and electronics (n = 16) sectors. Notably, 30% of these supportive industry submissions simultaneously requested specific exemptions for critical applications. This hybrid stance frequently utilized societal and environmental arguments, might indicate that a segment of the industry is actively transitioning toward PFAS free alternatives but requires specific legal derogations to manage the technical transition phase. Furthermore, this limited industrial support may reflect strategic positioning by "early movers" who have already developed PFAS free alternatives and may gain a competitive advantage from stricter regulatory standards that raise market entry barriers for less prepared competitors. Crucially, the striking scarcity of supportive industry submissions suggests that the primary power imbalance in this regulatory space is not merely between civil society and corporate actors, but within industry itself. The lobbying and economic power of incumbent, PFAS dependent sectors appear to heavily drown out the voices of emerging, PFAS free innovators. Consequently, the AI-assisted large scale comment analysis clearly reveals the asymmetry in lobbying power by industries that rely heavily on PFAS functionality across manufacturing supply chains, as compared to that of other stakeholders.

4. Conclusion and implications for policy and transparency

The combined manual and AI-assisted large scale comment analysis framework reveal how the 2023 ECHA PFAS consultation became a stage for industrial lobbying. Rather than overt opposition, industry actors, across sectors and countries, strategically mobilized a set of strategic rhetorical clusters identified in this study: primarily the lack of alternatives and concerns for economic consequences by losing competitiveness and impact on business. To a lesser extent, industry utilized societal and scientific arguments, such as the classification of specific PFAS as "polymers of low concern". Together, these rhetorical clusters collectively shift the narrative from one focused on chemical risks and public health, into one centered on European competitiveness and technological indispensability.

These findings highlight a significant governance challenge. Even within transparent regulatory processes, informational asymmetries persist when well resourced global actors dominate the consultation space. When mass participation is leveraged to "drown" the regulator in technical complexity, the process risks shifting from democratic deliberation to a venue for regulatory delay. This dynamic can amplify industry favoured perspectives, potentially marginalizing public health and environmental concerns (Coria et al., 2022). This dynamic directly strains ECHA's explicit mandate, which is to gather robust, objective socio-economic and technical data to help its scientific committees (Committee for Risk Assessment (RAC) and the Committee for Socio-Economic Analysis (SEAC)) draft balanced, enforceable regulations.

To support this institutional need and foster regulatory resilience, we

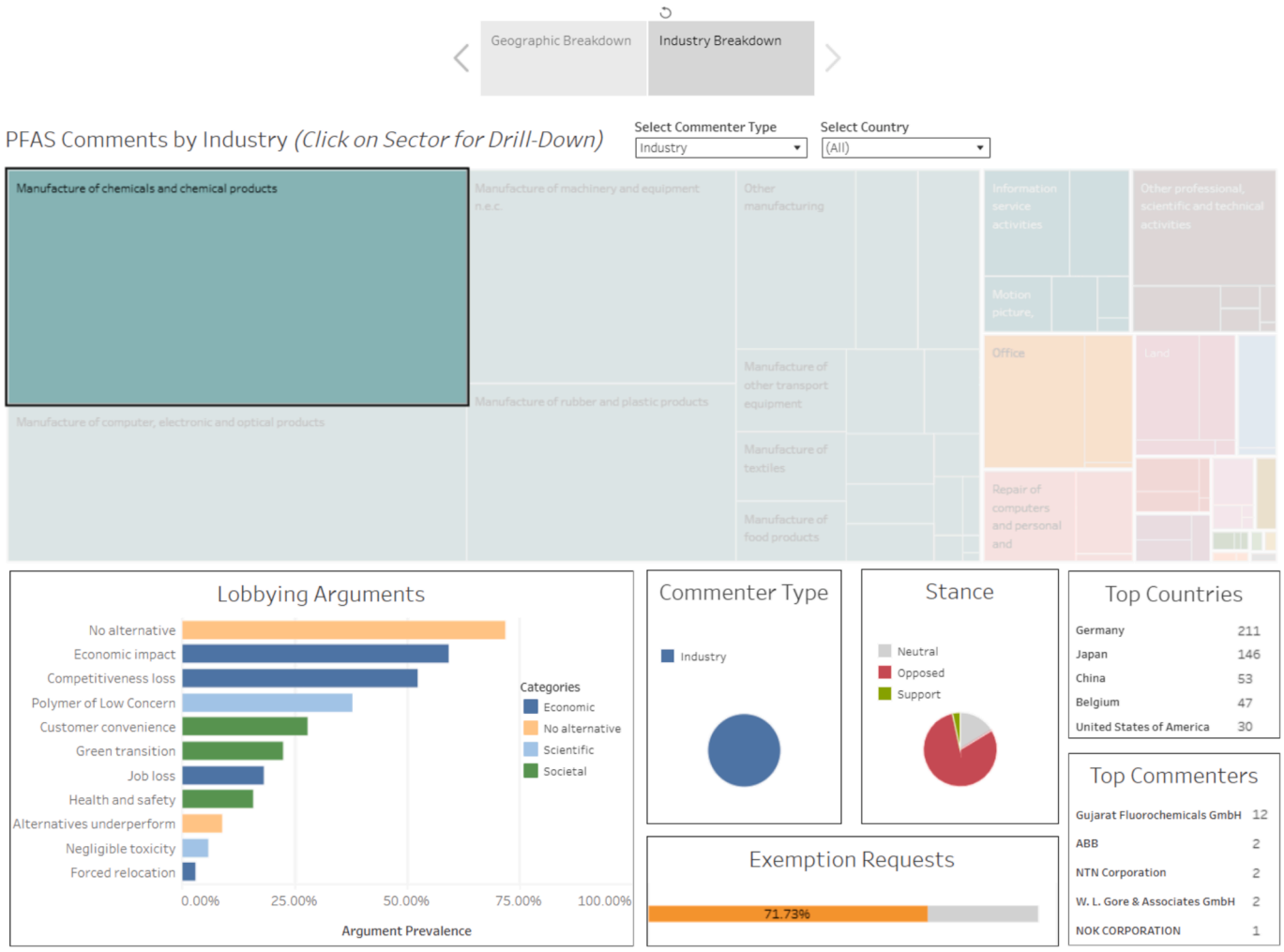


Fig. 4. Example of sector level analysis for the manufacturers of chemicals and chemical products. The tree map (top) shows the relative contribution of different industries, while the lower panels display lobbying arguments, stance on restriction, and exemption requests for this sector. The dashboard also identifies dominant geographical origins and top volume commenters. The interactive version of analysis is.
available at: https://public.tableau.com/app/profile/marko.poldma/viz/Dashboard22may/Story1

argue that systematic mapping of these lobbying patterns must become a standard institutional practice. While bodies like ECHA expected to focus strictly on technical data, the logistical hurdle lies in the sheer quantity of thousands of submissions. The integration of AI-assisted tools and open-access dashboards, as demonstrated in this study, provides a mechanism to handle this massive dataset, not by replacing expert evaluation, but by clustering specific claims such as 'no alternative' to quickly isolate widespread, coordinated industry narratives from genuine, unique technical bottlenecks where alternatives truly do not yet exist. Ultimately, ensuring that consultation mechanisms serve democratic deliberation and are inclusive towards the voices of citizens and consumers, requires sustained interdisciplinary engagement between data science, environmental governance, and public ethics.

CRediT authorship contribution statement

Mohammad Sadia: Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Nathalia Bastos:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Janne van Asselt:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Thomas Timmerman:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Marko Põldma:** Writing – review & editing, Validation, Software, Investigation, Formal analysis, Data curation. **Annemarie van Wezel:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.

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Data availability

Data will be made available on request.

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